🚀 HACKATHON PITCH

Web / Open Source / Focus Companion

Liread

Small daily pages. Massive compounding wisdom.

A calm, distraction-free reading sanctuary with deep-focus timers, procedural ambient soundscapes, offline reading, and mindful community accountability.

Creator: Muluken Tsegaye • **Platform:** Progressive Web App & Cloud Sanctuary

`https://liread.vercel.app` • `Production Ready`

The Modern Reading Dilemma

Why deep reading is collapsing in an era of algorithmic distraction

144m

DAILY ALGORITHMIC SOCIAL SCREEN TIME

Average adult spends 2.4+ hours daily in fragmented feeds.

< 14m

DAILY TIME DEVOTED TO DEEP READING

Literary contemplation has plummeted by 40% over the last decade.

74%

ABANDON BOOKS BEFORE THE HALFWAY MARK

Notification fatigue and lack of accountability break reading momentum.

⚡ Cognitive Fragmentation

Endless notifications and micro-stimuli rewire human attention spans, making sustained, reflective deep reading feel exhausting.

🏛️ Solitary Isolation

Traditional book reading is entirely isolated. There are no friendly, pressure-free accountability systems to sustain daily consistency.

📦 Siloed Marginalia

Brilliant quotes and life-changing perspectives remain trapped in paper margins or proprietary e-readers rather than shared wisdom.

THE ANTIDOTE

Introducing Liread

A serene, mindful sanctuary designed to restore human contemplation

Where Deep Contemplation Meets Community Accountability

Liread is not another bloated social network or gamified tracker. It is a **calm reading sanctuary** engineered from the ground up to protect your attention, compound daily pages into masterworks, and celebrate quiet dedication.

✦ Distraction-Free Audio Engine

✦ Zero-Algorithm Public Feed

✦ Real Relational Likes

The Compounding Habit Engine

Reading just **15 minutes every morning** compounds into:

18 Books
COMPLETED MASTERWORKS / YEAR

109 Hours
QUIET, NOTIFICATION-FREE FOCUS

1. Focused Immersion

Built-in Pomodoro & open reading timers paired with client-side synthesized ambient white noise, rain, and fireplace sounds.

2. Zero-Friction Library

Instant access to millions of classic and public domain masterworks via Open Library & Project Gutenberg with offline reader.

3. Mindful Commonplace

Capture marginalia, quote gems, and takeaways. Real upvotes from real human readers without fake bots or algorithmic doom loops.

Core Product Capabilities

Key interactive features built into the Liread application

⏱️ Distraction-Free Timer

Full-screen HUD timer with customizable daily goals (15m, 30m, 60m), real-time progress arcs, and automatic streak incrementation.

- Floating bottom timer sync
- Daily streak calculation triggers

🎧 Procedural Ambient Audio

Zero audio files downloaded. Pure Web Audio API oscillators synthesize authentic rain, fireplace crackles, deep space, and birdsong.

- 100% offline synthesis
- Real-time frequency visualizer

📖 ReadEra Offline Engine

Browser-native EPUB and PDF reader powered by IndexedDB. Upload any personal document and read offline anywhere.

- EPUB flow rendering + PDF.js
- Persistent reading bookmark & zoom

🌐 Dynamic Catalog & Spotlights

Integrated Open Library API search across 14 curated genres, automated book cover rendering, and weekly algorithmic book spotlights.

- Dual in-memory / session caching
- Instant 0ms public domain downloads

💬 Social Commonplace Feed

Share inspiring book passages and personal reflections. Strictly real upvotes and comments with explicit zero states ("No likes yet").

- One-click passage bookmarking
- Rich reader profiles & follow network

🕯️ Live Co-Reading Sanctuary

Real-time Supabase Presence channels show fellow readers immersed in books worldwide, fostering quiet solidarity without pressure.

- Real-time presence heartbeat
- Daily & weekly reading leaderboards

SYSTEM ENGINEERING

Technical Architecture

Modern, responsive, and type-safe full-stack web architecture

CLIENT LAYER (REACT 18 + VITE)

React 18 & TypeScript
Strict type safety, zero compile warnings, component modularity.

Zustand State Store
Lightweight stores (`authStore`, `feedStore`, `timerStore`, `readerStore`) with persistent middleware.

Tailwind CSS & Tokens
Fluid responsive layout, dark/light theme variables, custom typography.

Web Audio & Canvas Engine
Procedural ambient sound generator and dynamic clothbound cover canvas renderer.



EDGE & API GATEWAY

Vercel Serverless Functions
`/api/book-file.ts`: Streaming public domain EPUB/PDF CDN proxy with fallback resolution.

Open Library REST APIs
Querying `/subjects`, `/works`, and `/search` with debounced abort controllers.

Telegram Bot Gateway
`/api/telegram.ts`: Webhook bridge for community book deliveries.

IndexedDB (ReadEra Storage)
Client-side offline document caching for zero-latency reader loads.



BACKEND (SUPABASE CLOUD)

PostgreSQL 15 Database
Relational schema with foreign keys, cascade deletes, and automated triggers.

Supabase Auth & RLS
Row-level security policies guarding user reads/writes with zero-leak protection.

Realtime Engine (WebSockets)
Live presence for co-reading sanctuary and instant post upvote sync.

PL/pgSQL Functions (RPC)
Atomic streak computation and leaderboard window ranking (`DENSE_RANK()`).

⚡ **Sub-Second Cold Starts:** Vite asset splitting generates chunked bundles under 125kB gzip.

🔒 **Zero-Leak Security:** No hardcoded secrets, client-side sanitized rate-limit notices, strict RLS.

📱 **PWA & Offline First:** Standalone installable web app capable of reading books without internet.

Database Architecture & Schema

PostgreSQL schema with atomic triggers, RLS policies, and relational integrity

profiles — Core User

id (PK)	UUID -> auth.users
username	TEXT UNIQUE
full_name	TEXT
total_minutes	INTEGER DEFAULT 0
current_streak	INTEGER DEFAULT 0
rank_tier	TEXT (Novice->Master)

posts — Marginalia

id (PK)	UUID DEFAULT gen_random
user_id (FK)	UUID -> profiles.id
book_title	TEXT NOT NULL
quote_text	TEXT
upvotes_count	INTEGER (Synced)
comments_count	INTEGER (Synced)

reading_sessions — Time Tracking

id (PK)	UUID DEFAULT gen_random
user_id (FK)	UUID -> profiles.id
book_title	TEXT NOT NULL
duration_mins	INTEGER NOT NULL
session_date	DATE NOT NULL
created_at	TIMESTAMPTZ

post_upvotes — Relational Likes

id (PK)	UUID
post_id (FK)	UUID -> posts.id
user_id (FK)	UUID -> profiles.id
UNIQUE	(post_id, user_id)

Enforces one upvote per user.

bookshelf — Personal Shelf

id (PK)	UUID
user_id (FK)	UUID -> profiles.id
title & author	TEXT NOT NULL
status	want_to_read | reading | done
progress_pct	INTEGER DEFAULT 0

saved_posts — Stash

id (PK)	UUID
user_id (FK)	UUID -> profiles.id
post_id (FK)	UUID -> posts.id
UNIQUE	(user_id, post_id)

Private bookmarked passage stash.

Automated Triggers: `on_reading_session_created` auto-computes streaks & upgrades rank tiers. Upvote & comment changes atomically sync counter caches.

RLS: Strictly Enforced

STRATEGIC EVOLUTION

Future Roadmap

Three high-impact horizons expanding the Liread ecosystem

HORIZON 1 — Q1 2027

🤖 AI Contextual Marginalia & Semantic Search

Context-aware AI companion running on user-directed prompts to demystify complex philosophy, define archaic terms, and surface connections across saved passages without leaving the page.

- Zero-spoiler chapter synthesis
- Vector embedding search across your marginalia
- Etymology & philosophical background cards

HORIZON 2 — Q2 2027

👥 Asynchronous Micro-Clubs & Shared Margins

Collaborative reading circles where book clubs read the same masterwork simultaneously. Readers can reveal each other's margins and audio voice notes strictly after finishing the chapter.

- Spoiler-safe chapter gates
- Voice memo marginalia attachments
- Private book circles for friend groups & classes

HORIZON 3 — Q3 2027

📱 Native Mobile & E-Ink Hardware Sync

Direct two-way synchronization with dedicated e-readers (Kindle, Kobo, Boox) and high-contrast e-ink web companion mode, alongside native iOS/Android offline widgets.

- Bi-directional Kindle highlight imports
- Ultra-low power monochrome E-Ink layout mode
- Lock-screen reading streak widgets

THE LIREAD MISSION

"Reading is not dying — it is simply waiting for a *calm, distraction-free sanctuary* to call home."

By fusing deep-focus ambient technology, open literary access, and mindful human accountability, **Liread** empowers readers worldwide to reclaim their attention and build lifetime compounding wisdom.

Live Web Application: liread.vercel.app

Open Source GitHub: mulukentsegaye/liread-web

Thank you for your time & consideration. Let's make the world read again.